

claude-windows / 985cc286-9e1e-4217-87c3-2d418bc4fa9c

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\985cc286-9e1e-4217-87c3-2d418bc4fa9c\subagents\workflows\wf_98c1f280-b9f\agent-ae07d7a98b3248063.jsonl`  
- SHA-256: `f719eadd67cfb8da3e9d0c422cd58abb75789f399f19d085bdf0380097ae90b1`  
- Source modified: `2026-06-14T06:37:43+00:00`  
- Imported at: `2026-07-05T16:48:25+00:00`  
- project: `wf_98c1f280-b9f`  
- session_id: `985cc286-9e1e-4217-87c3-2d418bc4fa9c`
```

Transcript

1. user (2026-06-14T06:37:27.290Z)

Write a tight markdown report for the repo owner. Inputs:

FINDINGS:

```
[  
  {  
    "component": "PERSON cue (person_detector.py producer + fusion_features.py per-window  
feature math, wired through FusionSegmenter and the fusion_golden/fusion_compare LOVO  
measurement)",  
    "present": true,  
    "default_off": true,  
    "no_special_casing": true,  
    "retained": true,  
    "evidence": [  
      "PRESENT (producer): backend/pipeline/detectors/person_detector.py:50 `class  
PersonDetector` ? torchvision detector + ByteTrack + motion windowing; :140  
`extract_action_windows_from_chunk`, :258 `detections_per_frame`, :305 `detections_over_windows`  
(single-pass per-window detections feeding fusion).",  
      "PRESENT (feature math): backend/pipeline/detectors/fusion_features.py defines all 5  
features ? :97 `players_in_court`, :102 `in_court_ratio`, :109 `mean_player_motion`, :130  
`two_sided_motion`, :168 `shuttle_player_colocation`, aggregated by :200  
`compute_person_features`; :194 `PERSON_FEATURE_NAMES = ['players_in_court', 'two_sided_motion',  
mean_player_motion', 'in_court_ratio', 'shuttle_player_colocation']`. Pure (no torch/cv2/I/O),  
scale/translation-invariant by design.",  
      "DEFAULT-OFF (cue flag): backend/config/models.py:96 `cue_flags: dict[str, bool] =  
Field(default_factory=dict)` (empty ? person OFF); models.py:94-95 comment 'Person cue is OFF  
unless explicitly enabled'. tests/test_config.py:48 `assert f.cue_flags == {}` # person cue  
OFF`.",  
      "DEFAULT-OFF (segmenter): backend/config/models.py:112 `default_segmenter: str =  
'yolo_hybrid'` and config.json:26 `\"default_segmenter\": \"gemini\"` ? neither is the  
`fusion_hybrid` segmenter. FusionSegmenter is opt-in: fusion_hybrid.py:23-26 'Registered as  
fusion_hybrid, NOT the default segmenter ? it is opt-in. Promotion to default happens only on  
measured LOVO lift? never silently.",  
      "DEFAULT-OFF (enrichment gated on flag): backend/pipeline/segmenters/fusion_hybrid.py:88  
`if fcfg.get('cue_flags', {}).get('person', False):` guards `_person_features`;  
fusion_hybrid.py:109-112 lazily builds PersonDetector / loads torchvision ONLY when the flag is  
on, so the default path never touches the GPU.",  
      "DEFAULT-OFF (served gate is a separate knob, also default 0):  
backend/config/models.py:102 `min_players: int = Field(default=0, ge=0)`; fusion_hybrid.py:148  
`if person_on and min_players > 0 and st.get('players_in_court',0.0) < min_players: continue` ?  
only fires when the operator both enables person AND raises min_players>0. Default thresholds  
keep every window (fusion_hybrid.py:136-138).",  
      "NO SPECIAL-CASING (features are learned columns): backend/eval/fusion_compare.py:6 'the  
person features are extra columns the learned LogisticRegression weights';  
fusion_compare.py:57-61 builds two Variants differing ONLY by `feature_names` ? shuttle_only vs  
`SHUTTLE_FEATURE_NAMES + PERSON_FEATURE_NAMES`. experiment.py:234
```

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`model.predict_proba(_feature_matrix(vc, variant.feature_names))` ? a LogisticRegression
(experiment.py:49,293 `LogisticRegression().fit(X,y,variant.feature_names)` scores them; there
is NO hard if/else on any person feature in the decision path.",
    "NO SPECIAL-CASING (producer?feature?scorer chain): backend/eval/fusion_golden.py:108
`ff.compute_person_features(...)` produces the columns per candidate; :112-114 persists them as
`person` dict next to `shuttle`; fusion_compare.py:47-48 loads them into `features[fn]`; the
only hard gate (`min_players`) is the eval-side Gate B (experiment.py:200-210 `_gate_mask`)
which is a SEPARATE default-0 knob, not the path by which the cue enters the learned decision.",
    "RETAINED (code intact post-measurement): n=6 result is non-significant ?
eval_baselines/experiment_leaderboard.json shows shuttle_only F1=0.307 vs shuttle_person_fusion
F1=0.286, delta f1=-0.0205, honest_delta_f1 ci_low=-0.165 ci_high=+0.162 ci_excludes_zero=false,
sign_test 2W/4L p=0.6875. Despite this, person_detector.py and fusion_features.py are unchanged
and fully functional.",
    "RETAINED (only a docs log, no code change): branch docs/i3-fusion-measured commit 9eb06f4
`docs(quality): log fusion P3+P4 measured ?F1` touches ONLY docs/QUALITY_ITERATIONS.md (+19
lines); `git diff --stat origin/master...HEAD` = `docs/QUALITY_ITERATIONS.md | 19 +++` and
nothing else. Commit message: 'neither cue is promotable; both stay default-OFF (promote-only-
on-lift)? Negative/uncertain results kept on purpose ? the honest ablation backbone.'",
    "RETAINED (re-measurable): fusion_golden.py writes replayable
`<name>_golden_features.json` per video (the persisted feature substrate,
fusion_golden.py:13-15) so `fusion_compare` can re-run forever without re-detecting;
FusionSegmenter persists the full candidate superset regardless of gating
(fusion_hybrid.py:18-20) so the harness can re-score any variant as the corpus grows."
],
    "concerns": [
        "Feature-quality caveat acknowledged in code/docs: the n=6 run was extracted with
`court_polygon=None` (fusion_golden.py:109 passes `court_polygon=None`), so `players_in_court`
and `two_sided_motion` count spectators / adjacent-court people. The commit msg notes 'Person
adds noise (no court polygons -> players_in_court counts spectators)'. This means the non-
significant/negative ?F1 partly reflects un-tuned features, not necessarily a useless cue ? re-
measuring with court polygons (FusionConfig.court_polygon, models.py:104) is the obvious next
experiment before any conclusion. Not a guardrail violation, but the negative result should not
be read as 'person cue doesn't work'.",
        "The person cue's served Gate B (`min_players`, fusion_hybrid.py:148; experiment.py
_gate_mask:207-209) IS a hard threshold if/else ? but it is a SEPARATE operator-set decision-
gate (default 0 = off), distinct from how the feature enters the learned scorer. The 'no
special-casing' criterion holds for the learned decision path (feature columns), but reviewers
should note `players_in_court` does also have a hard-gate consumer; both are default-OFF so
neither runs by default.",
        "`compute_person_features` is invoked from TWO call sites with slightly different framing:
fusion_golden.py (eval, court_polygon=None) and fusion_hybrid.py:130 (served, court_polygon from
config). Production wiring (fusion_hybrid._person_features) is explicitly only mock-tested ?
fusion_hybrid.py:108 '? real-video verification is owner-side; this wiring is mock-tested.' So
the served person path has not been exercised on real video; only the offline eval path produced
the n=6 numbers. Re-measurability claim is sound, but the served path's correctness on real
footage is unverified here (GPU-free container).",
        "n=6 is too small to conclude anything either direction ? both the audit's 'non-
significant' framing and the code comments agree. The retention decision is correct precisely
because the result is inconclusive, not because the cue was shown harmful."
    ]
},
{
    "component": "audio cue (audio_features.py + hit_detector.py, consumed by
eval/fusion_audio.py)",
    "present": true,
    "default_off": true,
    "no_special_casing": true,
    "retained": true,
    "evidence": [
        "PRESENT producer: backend/pipeline/audio/hit_detector.py:136 `def
detect_hits_in_video(video_path, sr=16000, **kw) -> List[HitEvent]` ? full onset chain
extract_audio->load_wav->onset_envelope->moving_stats->detect_hits (adaptive mean+k*std
threshold, lines 28-133).",
        "PRESENT feature math: backend/pipeline/detectors/audio_features.py:15
`AUDIO_FEATURE_NAMES = [\"audio_hit_count\", \"audio_hit_rate\", \"audio_hit_strength_mean\"]`;

```

```

:18 `def compute_audio_features(hits, window_start, window_end) -> Dict[str,float]` returns a
numeric dict (0.0 on empty window, :28). Unit-tested in tests/test_audio_features.py:16-33.",
    "DEFAULT_OFF (served default): backend/config/models.py:112 `default_segmenter: str =
`yolo_hybrid`` and config.json:26 `\"default_segmenter\": \"gemini\"` ? never 'fusion'.
Selection is `req.segmenter or default_segmenter` (backend/main.py:712, docs/CODE_MAP.md:31).
Audio is unreachable from any served segmenter.",
    "DEFAULT_OFF (audio is eval-only): the ONLY importers of
detect_hits_in_video/compute_audio_features are backend/eval/fusion_audio.py:24-25 and
backend/eval/audio/e1_probe.py:18. `git grep` for these imports excluding backend/eval and tests
returns EMPTY ? zero production importers.",
    "DEFAULT_OFF (not even in the opt-in fusion segmenter):
backend/pipeline/segmenters/fusion_hybrid.py `_enrich_candidate_stats` (:76-90) persists only
`net_crossings` (:85) and, behind `cue_flags['person']` (:88), person features ? it NEVER calls
compute_audio_features or detect_hits_in_video. There is no `cue_flags['audio']` in FusionConfig
(models.py:78-108).",
    "NO_SPECIAL_CASING: audio enters the decision purely as feature columns the learned
LogisticRegression weights. backend/eval/fusion_audio.py:84 builds the variant with
`feature_names = SHUTTLE + PERSON + af.AUDIO_FEATURE_NAMES`; experiment.predict
(experiment.py:227-238) does `model.predict_proba(_feature_matrix(vc, variant.feature_names))`
then thresholds the probability. No `if audio_hit_count > X` branch exists; `_gate_mask
(experiment.py:200-210) only references net_crossings/players_in_court, never audio.",
    "NO_SPECIAL_CASING (per-window math is pure numeric): audio_features.py:30-33 returns
{audio_hit_count: float(n), audio_hit_rate: n/dur, audio_hit_strength_mean: mean(strength)} ?
values, never a hard-coded keep/drop decision.",
    "RETAINED (no deletion): `git log --all --diff-filter=D` over both files + fusion_audio.py
returns EMPTY (no commit ever removed them). Both still on master; added in #122 dc19bf2 'fusion
P4 ? audio cue + incremental-lift measurement (default-OFF)'.",
    "RETAINED (kept despite non-significant n=6): docs/QUALITY_ITERATIONS.md:52 records `+
person + audio (P4 incr.) | 0.366 | +0.079 (+27.7%) | [-0.054, +0.203] | 4W/2L (p=0.688) | CI
clears 0: No`; :54-60 conclude `both stay default-OFF` and `:59-60 `Negative/uncertain results
are kept on purpose ? this is the honest ablation backbone`. Re-measurable as N grows via
`fusion_audio.py --augment --compare` (the producer + feature columns persist into
*_golden_features.json).",
    ],
    "concerns": [
        "The real-audio producer path (detect_hits_in_video: ffmpeg extraction + numpy onset
detection) has NO offline test ? this dev container lacks ffmpeg/numpy/cv2, so only the pure
compute_audio_features math and the synthetic LOVO driver are covered
(tests/test_audio_features.py). The actual hit-detection quality on real video is owner/GPU-
side-verified only; the n=6 ?F1 came from real runs but is not reproducible in CI.",
        "Audio is NOT yet wired to ANY served segmenter, even the opt-in fusion_hybrid. Unlike
Gate A (which already has a served `min_crossings` knob and a persisted net_crossings stat) and
the person cue (which has `cue_flags['person']` + a `_person_features` enrichment seam), there
is NO `cue_flags['audio']`, no audio enrichment seam in fusion_hybrid._enrich_candidate_stats,
and audio features are never persisted into candidate_segments.stats. So even if audio measured
promotable, it would require new production wiring ? it currently exists ONLY in the offline
eval harness. This is correct/honest for a default-OFF cue but worth flagging: 'present + re-
measurable' is true only in eval, not as a flag-flippable served path.",
        "The +0.079 audio lift is measured as an INCREMENTAL gain ON TOP of the person cue
(shuttle+person vs shuttle+person+audio, fusion_audio.py:79-86), and person itself was a -0.021
regression. QUALITY_ITERATIONS.md:58 itself notes audio should next be measured on shuttle-only
directly 'without the person handicap' ? so the headline number is confounded by the person
baseline and is not yet a clean audio-vs-shuttle measurement."
    ]
},
{
    "component": "COURT BOUNDS ? court-polygon person-occupancy signal
(fusion_features.point_in_polygon / person_in_court / in_court_counts / players_in_court /
in_court_ratio, wired via FusionSegmenter, gated by FusionConfig.court_polygon)",
    "present": true,
    "default_off": true,
    "no_special_casing": true,
    "retained": true,
    "evidence": [
        "MATH PRESENT: backend/pipeline/detectors/fusion_features.py:30-48 `point_in_polygon`

```

(ray-casting, <3 verts => False); :51-57 `\_foot\_norm` (bottom-centre foot point = court-membership anchor); :67-76 `person\_in\_court` ? 'if court_polygon is None: return True' (count-everyone) else foot-point-in-polygon; :79-86 `in\_court\_counts` per-frame in-court count; :97-99 `players\_in\_court` median; :102-106 `in\_court\_ratio` fraction of frames >= min_players. Pure, no torch/cv2.",

"CATALYST ACTIVATION: backend/pipeline/detectors/fusion_features.py:200-215 `compute\_person\_features(..., court\_polygon=None, ...)` threads the polygon into `in\_court\_counts` (:207) and `two\_sided\_motion` (:210). With None it degrades to 'every detection counts'; with a polygon every count/ratio/two-sided value changes ? proven by tests/test_fusion_features.py:54-67 (`in\_court\_counts(..., LEFT\_HALF)==[1,1,0,2]` vs `(...,None)==[2,1,0,2]`; players_in_court 1.0; in_court_ratio 0.25), :38-49 (person_in_court LEFT_HALF True/False vs None always True), :94-101 (two_sided_motion 1.0 with None vs 0.0 with LEFT_HALF). So the math activates the moment a polygon is supplied.",

"PRODUCTION WIRING activates on config: backend/pipeline/segmenters/fusion_hybrid.py:126-131 ? `poly = fcfg.get('court\_polygon')`; `polygon = [(float(p[0]),float(p[1])) for p in poly] if poly else None`; passed into `ff.compute\_person\_features(..., court\_polygon=polygon, net=net)`. So once `indexing.fusion.court\_polygon` is set (e.g. when Roora labels arrive) the masked path engages with no code change ? the catalyst-later mechanism.",

"DEFAULT-OFF (config): backend/config/models.py:104 `court\_polygon: Optional[list[list[float]]] = None`; :96 `cue\_flags: dict[str,bool] = Field(default\_factory=dict)` (person cue absent => OFF); :99-102 `min\_crossings`/`min\_players` default 0 (=OFF). FusionConfig docstring :85-86 confirms None => Gate B counts every detection (player-count-only).",

"DEFAULT-OFF (served path): backend/config/models.py:112 `default\_segmenter: str = 'yolo\_hybrid'` and config.json:26 ships `gemini` ? neither is `fusion\_hybrid`, so the served path never instantiates FusionSegmenter by default. fusion_hybrid.py:88 only computes person features `if fcfg.get('cue\_flags',{}).get('person',False)` (default False) ? so the court-polygon code never even runs in the default path. fusion_hybrid.py:22-26 docstring: persists but gates nothing by default; 'Registered as fusion_hybrid, NOT the default segmenter ? it is opt-in.'",

"DEFAULT-OFF (eval): backend/eval/fusion_golden.py:108-109 calls `ff.compute\_person\_features(..., court\_polygon=None, ...)` ? explicitly None today (the 'currently None' the prompt notes); the degrade-to-count-everyone path, re-runnable with a polygon later.",

"NO SPECIAL-CASING: the court-derived person features enter the DECISION only as learned-scorer columns. backend/eval/fusion_compare.py:47-48 loads `players\_in\_court`/`in\_court\_ratio`/etc. into the feature dict; :59-61 the fusion Variant differs from shuttle_only ONLY by `feature\_names = SHUTTLE + PERSON`; the LogisticRegression (distill_local.py:116/135 `LogisticRegression().fit(X,y,FEATURE\_NAMES)`) weights them. fusion_compare.py docstring :5-7: 'the person features are extra columns the learned LogisticRegression weights ... not a hunch.' The ONLY if/else on a court feature is the OPTIONAL served Gate B (fusion_hybrid.py:148 `if person\_on and min\_players>0 and st['players\_in\_court']<min\_players`), which is a config-gated threshold defaulting OFF (min_players=0), NOT a hard-coded decision branch.",

"RETAINED despite non-significant n=6: git log shows only feature-ADD commits on these files (cbf2236 P2, 39e961b P3, d7b1f51 honest ?F1) ? no deletion/disable commit after the result. docs/QUALITY_ITERATIONS.md:51 records +person ?F1 ?0.021, CI [?0.165,+0.162], 2W/4L p=0.688, 'CI clears 0: No'; :54-55 'both stay default-OFF (promote-only-on-lift)'; :55 explicitly attributes the miss to 'no court polygons ? players_in_court counts spectators'; :59-60 'Negative/uncertain results are kept on purpose ? this is the honest ablation backbone'. Producer (fusion_golden.py) + feature math (fusion_features.py) + scorer columns (fusion_compare.py) all still exist and are re-measurable when the corpus grows. docs/NEXT_STEPS.md:71-72 keeps 'Court mask = optional fusion.court_polygon' as a live capability."

],

"concerns": [

"No real-video activation of the polygon path has been run: fusion_hybrid.py:108 flags person-feature wiring as 'real-video verification is owner-side; this wiring is mock-tested', and CLAUDE.md notes the dev container has no GPU/torch/cv2. The polygon MATH is unit-tested with LEFT_HALF, but the end-to-end masked person-detection pass on actual video (and thus the catalyst's real-world lift) is unverified ? by design, owner-gated.",

"No court_polygon is committed anywhere yet (config.json has no fusion block; FusionConfig default None; fusion_golden passes None literally). The catalyst is fully plumbed but dormant ? it only fires after someone supplies the polygon (the prompt's 'once Roora labels arrive')."

There is no producer that auto-derives a polygon from court detection today; it must be hand/label-supplied via config.",

"Coordinate-space coupling risk for the catalyst: polygons must be normalised 0-1 and `two\_sided\_motion`/foot-points use per-axis normalisation (x/width, y/height) ? a future Roora-supplied polygon in pixel space or a different convention would silently mis-mask (point_in_polygon returns plausible-but-wrong booleans, not an error). No validation that court_polygon vertices are in [0,1] exists in FusionConfig (only `Optional[list[list[float]]`); a malformed polygon degrades silently rather than failing loudly.",

"Minor: the n=6 person result is reported as a COMBINED person-feature-set ?F1; the court-occupancy sub-signal's individual contribution is not separately measured, so 'court bounds specifically helps' remains untested even setting aside the None-polygon handicap. Re-measuring per-feature once polygons exist would be the honest next step."

```
]
},
{
  "component": "FusionSegmenter + config defaults
(backend/pipeline/segmenters/fusion_hybrid.py, trajectory_hybrid.py, backend/config/models.py)",
  "present": true,
  "default_off": true,
  "no_special_casing": true,
  "retained": true,
  "evidence": [
    "PRESENT: backend/pipeline/segmenters/fusion_hybrid.py:39 `class
FusionSegmenter(TrajectoryHybridSegmenter):` with `family = \"fusion\"` (:42) and `name ->
\"fusion_hybrid\"` (:56); registered at :154 `segmenter_registry.register(FusionSegmenter)`.
Producers exist and are git-tracked: backend/pipeline/detectors/fusion_features.py
(PERSON_FEATURE_NAMES def at :194), rally_gate.py (Gate A net-crossing).",
    "PRESENT (config): backend/config/models.py:78 `class FusionConfig` wired into
IndexingConfig:142 `fusion: FusionConfig = Field(default_factory=FusionConfig)`.",
    "DEFAULT_OFF (segmenter): model default is NOT fusion ? models.py:112 `default_segmenter:
str = \"yolo_hybrid\"`; shipped config.json:26 `\"default_segmenter\": \"gemini\"`. Neither is
fusion. The only literal `\"fusion_hybrid\"` in backend/ is the segmenter's own name property
(fusion_hybrid.py:56) ? never assigned as a default. Module docstring fusion_hybrid.py:24-26
states it is 'Registered as fusion_hybrid, NOT the default segmenter ? it is opt-in.',
    "DEFAULT_OFF (cues): FusionConfig.cue_flags defaults empty -> person OFF. models.py:96
`cue_flags: dict[str, bool] = Field(default_factory=dict)`; gate read at fusion_hybrid.py:88 `if
fcfg.get(\"cue_flags\", {}).get(\"person\", False):` (default False ? no person detector / no
torch model loaded, :110-112 lazy import only when ON).",
    "DEFAULT_OFF (served gates no-op): models.py:99 `min_crossings: int = Field(default=0,
ge=0)` and :102 `min_players: int = Field(default=0, ge=0)`. fusion_hybrid.py:146-150 only drops
windows when `min_crossings > 0` / `min_players > 0`; at defaults `_select_for_handoff` keeps
every window ? docstring :136-138 'Default thresholds (0) keep every window ? identical to the
substrate.' Base class confirms identity: trajectory_hybrid.py:104 `return
list(range(len(windows)))`.",
    "NO_SPECIAL_CASING: the cue enters the DECISION as a feature COLUMN a learned
LogisticRegression weights, not a hard-coded decision branch. backend/eval/fusion_compare.py:5-7
'the two variants differ ONLY by feature_names ... the person features are extra columns the
learned LogisticRegression weights'. fusion_compare.py:57-61 builds variants whose only
difference is `feature_names=SHUTTLE_FEATURE_NAMES` vs `+ PERSON_FEATURE_NAMES`.
experiment.py:227-238 `predict()` runs `model.predict_proba(_feature_matrix(vc,
variant.feature_names))` then thresholds the learned probability; experiment.py:281-293 `_train`
fits `LogisticRegression().fit(X, y, variant.feature_names)`. `net_crossings` is a learned
shuttle feature: distill_local.py:40 `FEATURE_NAMES =
[\"duration\", \"peak_velocity\", \"mean_velocity\", \"active_ratio\", \"net_crossings\"]`, re-used
as SHUTTLE_FEATURE_NAMES (fusion_golden.py:33).",
    "RETAINED: nothing was deleted/disabled despite a non-significant n=6 result.
eval_baselines/experiment_leaderboard.json (n=6) records shuttle_only vs shuttle_person_fusion
`delta_f1 = -0.0205`, `honest_delta_f1.ci_excludes_zero: false`, sign_test 2W/4L p=0.6875 ? a
clear non-significant/negative lift. Yet the producer + feature math remain: fusion_hybrid.py:85
still computes `stats[\"net_crossings\"]`, :88-89 still computes person features,
fusion_features.py/rally_gate.py/fusion_golden.py/fusion_compare.py are all still git-tracked
and importable. Re-measurable when the corpus grows (load_videos just re-reads
*_golden_features.json).",
    "PERSISTENCE INVARIANT (supports retained/re-measurable): even when served gates ARE
raised, persisted candidates stay the full superset ? fusion_hybrid.py:20 and :136-138
```

'Persisted candidates are untouched (full superset)'; trajectory_hybrid.py:208-217 persists every window's stats unconditionally; only `\_select\_for\_handoff` indices (:225) are gated for the served handoff."

```

],
"concerns": [
    "The served Gate A/B path in fusion_hybrid.py:_select_for_handoff IS a hard if/else threshold branch (`min_crossings > 0 ... < min_crossings`, :146-150), NOT a learned scorer. This is fine because (a) it is default-OFF (thresholds 0) so it never fires on the served path, and (b) the no_special_casing criterion is about how the cue enters the LEARNED DECISION (the eval harness uses feature_names columns, which it does). But note the served production segmenter, if ever switched to fusion_hybrid with thresholds raised, would use hard-coded if/else gating ? not the LogisticRegression. The learned-scorer treatment lives only in the offline eval harness (backend/eval/experiment.py + fusion_compare.py), not in the served fusion_hybrid path. Promotion-to-default requires owner sign-off per docstring :24-26.",
    "Two distinct gate semantics share names: experiment.py:_gate_mask (:200-210) ALSO implements min_crossings/min_players as hard thresholds on the eval side, independent of the learned classifier ? so even in eval those two knobs are hard gates layered before the learned filter. The learned-column treatment applies to feature_names (incl. net_crossings/person features), while min_crossings/min_players are deliberately a separate declarative hard gate. This is intentional (Variant docstring experiment.py:90-96) but means 'net_crossings' can be consumed BOTH as a learned column and as a hard gate depending on which knob a variant sets.",
    "config.json ships default_segmenter='gemini' while the model default is 'yolo_hybrid' ? a known divergence (CODE_MAP.md:105). Neither is fusion, so default-OFF holds, but the served default is the cloud Gemini path, not yolo_hybrid; the fusion-vs-substrate equivalence claim is about wasb_hybrid, a third path. No impact on this component's default-OFF status.",
    "Real-video verification of the person path is owner-side only (fusion_hybrid.py:108 '? real-video verification is owner-side; this wiring is mock-tested'). The dev container lacks torch/cv2, so the person-cue producer is exercised only via mocks here ? re-measurability of the person columns at corpus growth depends on owner-side GPU runs, but the math/code is fully retained.",
    "A second leaderboard row (shuttle_person vs shuttle_person_audio) shows a positive +0.079 ?F1 for an AUDIO cue ? out of scope for this component but worth noting the audio variant is a separate eval driver (fusion_audio.py); it does not change the person/net-crossing default-OFF posture audited here."
]
},
{
    "component": "SCORING seam / no-special-casing (backend/eval/experiment.py ? Variant.feature_names, compare(), the learned LogisticRegression scorer)",
    "present": true,
    "default_off": true,
    "no_special_casing": true,
    "retained": true,
    "evidence": [
        "backend/eval/experiment.py:113-114 ? Variant carries `classifier_model: Optional[str] = None` and `feature_names: Optional[List[str]] = None`; a cue enters the learned decision purely by appearing in this list. Adding/removing a cue is a feature-LIST change, not a code edit.",
        "backend/eval/experiment.py:227-238 (predict) ? the ONLY learned-decision path: `if variant.feature_names: -> build `_feature_matrix(vc, variant.feature_names)` -> `model.predict_proba(...)` -> threshold the probability. The cue's name is data passed to the scorer; there is NO per-cue `if net_crossings.../if players...` branch in this path. `_feature_matrix` (195-197) and `_feature_column` (180-192) are name-driven and cue-agnostic (a missing column defaults to 0.0, line 186-188).",
        "backend/eval/classifier.py:40-73 (fit) + :75-79 (predict_proba) ? a generic numpy LogisticRegression over a feature matrix; it learns one WEIGHT per column (`self.w`), standardizes, sigmoids. The scorer has zero cue-specific logic ? every feature is just a column index. `feature_names` is stored as metadata only (line 48, 94).",
        "backend/eval/distill_local.py:40 ? `FEATURE_NAMES = [\"duration\", \"peak_velocity\", \"mean_velocity\", \"active_ratio\", \"net_crossings\"]` fed to `LogisticRegression().fit(X, y, FEATURE_NAMES)` (:116). PROOF the shuttle cue (`net_crossings`) already rides the learned scorer as a weighted COLUMN, not an if/else ? exactly the no-special-casing pattern. docs/MULTI_SIGNAL_FUSION_PLAN.md:128 confirms: `net_crossings` is already a feature'.",
        "Default-OFF in production: backend/config/models.py:112 `default_segmenter: str = \"yolo_hybrid\"` and shipped config.json:26 `\"default_segmenter\": \"gemini\"` ? neither is the fusion/learned path. FusionConfig (:96/:99/:102) `cue_flags={}`, `min_crossings=0`,

```

```

`min_players=0` all default-OFF. Grep for any non-eval import of
`experiment`/`LogisticRegression` across backend/ (excluding eval/) returned NO matches ? the
learned scorer never runs in the served pipeline by default.",
    "no-regression invariant test: tests/test_experiment.py:78-83
`test_disabled_cues_byte_identical_to_control` asserts a variant carrying only disabled cues
`predict(...) == predict(CONTROL, ...)` tests/test_experiment.py:129-133 shows the learned path
is a recipe (`Variant(feature_names=["x\\"])`) recovering F1 via LOVO ? a pure feature-list
config, no code branch.",
    "Retained after non-significant n=6 ?F1: memory current-state.md:12-17 records measured
honest LOVO (n=6): +person P3 ?F1=-0.021 (CI spans 0), +person+audio P4 ?F1=+0.079 (CI spans 0),
'neither promotable'. Despite that, the producers + feature math are all still present and re-
measurable: backend/pipeline/detectors/fusion_features.py (person feature math),
audio_features.py, backend/eval/fusion_golden.py/fusion_compare.py/fusion_audio.py, and the
experiment seam itself ? nothing deleted/disabled. fusion_hybrid.py:22-26 docstring: persisted
candidates remain the full superset for offline re-scoring; promotion 'happens only on measured
LOVO lift... never silently.'"
],
    "concerns": [
        "NOT strictly 'cues enter ONLY as feature columns'. The harness retains a SECOND,
deliberate decision path: hard-threshold Gate A/B. In the offline Variant, `_gate_mask`
(experiment.py:200-210) does `if variant.min_crossings > 0: ... col[i] >= variant.min_crossings`
and the same for `min_players` against `players_in_court` ? genuine if/else threshold branches
keyed to SPECIFIC feature names. The served segmenter mirrors this: fusion_hybrid.py:144-151
`_select_for_handoff` loops `if min_crossings > 0 and st.get('net_crossings',0.0) <
min_crossings: continue` / `if person_on and min_players > 0 and ... players_in_court <
min_players`. So the same cue (net_crossings) can enter the decision EITHER as a learned
weighted column (clean) OR via a hard-coded if/else gate. This is by-design
(MULTI_SIGNAL_FUSION_PLAN §3.1 'Decision-level gate, safe baseline, ships first, no labels' vs
§3.2 'Feature-level, primary, the learned path') and is default-OFF (thresholds 0), so it is NOT
a smuggled special-case ? but it means the literal claim 'never an if/else in the decision path'
is false. The gate is a parallel, label-free, hand-set knob, not part of the learned scorer.",
        "The measured + RECOMMENDED win is the if/else gate, not the learned column. current-
state.md:53/70 record Gate-A (net_crossings>=2) as the measured win (dF1 +0.074, 6/6, p=0.031)
with the recommendation `indexing.fusion.min_crossings=2`. So the path the project actually
proposes to promote is the hard threshold (§3.1), while the learned-scorer column for the same
signal (P3 person) measured non-significant. The no-special-casing learned seam exists and is
correct, but it is not currently the winning/served mechanism ? worth not overstating it as
'the' decision path.",
        "Gate-on-absent-feature silently maps to 0 and FAILS the gate (predict drops the window):
experiment.py:186-188 logs a warning then returns zeros, and test_experiment.py:111-114 asserts
an absent `net_crossings` -> 0 -> window dropped by a >0 gate. For the LEARNED path a zero
column is benign (just a feature value), but for the GATE path a missing persisted feature
silently turns a keep into a drop. Minor data-hygiene footgun, not a special-casing issue.",
        "spec_hash (experiment.py:148-153) includes feature_names/min_crossings/min_players, so a
recipe change is traceable ? good ? but the served production gate is driven by
IndexingConfig.fusion (config.json), a SEPARATE surface from the Variant. The offline Variant
and the served `_select_for_handoff` re-implement the same gate logic in two places
(experiment.py:200-210 vs fusion_hybrid.py:144-151); they currently agree but could drift since
there is no shared helper. Not a special-casing defect, but a duplication risk worth noting."
    ]
},
{
    "component": "fusion-pipeline files + experiment leaderboard on origin/master",
    "present": false,
    "default_off": true,
    "no_special_casing": true,
    "retained": true,
    "evidence": [
        "Branch/remote state: `git branch --show-current` => docs/i3-fusion-measured; `git log
origin/master -1` => 8836dfe chore: scripts/ ... (#138); `git rev-list --left-right --count
origin/master...HEAD` => 1\\t1 (diverged by one commit each side, nothing deleted).",
        "All 9 source files tracked (`git ls-files`):
backend/eval/{experiment,fusion_audio,fusion_compare,fusion_golden}.py,
backend/pipeline/audio/hit_detector.py,
backend/pipeline/detectors/{audio_features,fusion_features,person_detector}.py,

```

```

backend/pipeline/segmenters/fusion_hybrid.py.",
    "All 9 present in origin/master tree (`git ls-tree -r --name-only origin/master | grep
...`) ? each path returned plus its test (test_audio_features.py, test_fusion_compare.py,
test_fusion_features.py, test_fusion_hybrid.py, test_hit_detector.py,
test_person_detector.py).",
    "Per-file last commit on origin/master (`git log origin/master -1 -- <file>`):
person_detector.py=351a27e(#125); fusion_features.py=cbf2236(#117);
audio_features.py=dc19bf2(#122); hit_detector.py=df9cf80(#93); fusion_hybrid.py=cbf2236(#117);
fusion_golden.py=813d1f8(#126); fusion_compare.py=d7b1f51(#127); fusion_audio.py=d7b1f51(#127);
experiment.py=2e5892e(#130).",
    "eval_baselines/experiment_leaderboard.json is NOT tracked: `git ls-files` returns
nothing; `git cat-file -e origin/master:eval_baselines/experiment_leaderboard.json` => fatal:
'exists on disk, but not in origin/master'; same for HEAD; `git status --short` => `??
eval_baselines/experiment_leaderboard.json` (untracked, not gitignored ? `git check-ignore`
returned nothing).",
    "The leaderboard file DOES exist on disk (2506 bytes, mtime Jun 14 11:54) and DOES contain
both rows: P3 = variant_a `shuttle_only` (f1 0.3069) vs variant_b `shuttle_person_fusion` (f1
0.2863), delta f1 -0.0205; P4 = variant_a `shuttle_person` (f1 0.2863) vs variant_b
`shuttle_person_audio` (f1 0.3656), delta f1 +0.0792; both include honest_delta_f1 (bootstrap CI
+ sign test).",
    "Other eval_baselines/ files ARE tracked (gemini_wrapper_2026-06-10.json,
heuristic_lovo_n6.json) ? only the leaderboard JSON is the untracked exception; nothing under
eval_baselines was deleted."
],
    "concerns": [
        "present=false ONLY because eval_baselines/experiment_leaderboard.json is not committed to
git (untracked working-tree file). All 9 source/eval files ARE tracked and present on
origin/master, so if the leaderboard's git-tracking is not required, the source-file portion of
the check fully passes.",
        "The leaderboard JSON's content is correct and complete (both P3 and P4 fusion rows with
honest_delta_F1), it is simply not yet under version control ? recommend `git add
eval_baselines/experiment_leaderboard.json` and commit if it is meant to be a recorded baseline,
since the sibling eval_baselines JSONs are tracked.",
        "Current checkout is on branch docs/i3-fusion-measured (1 ahead / 1 behind origin/master)
? diverged, not fast-forwardable. Nothing was deleted, but local is not pointed straight at the
remote default branch HEAD.",
        "Measured results are honest: P3 fusion is a slight regression (delta f1 -0.0205, sign
test 2 wins/4 losses, CI includes zero); P4 audio is a positive lift (delta f1 +0.0792) but CI
still includes zero ? neither is statistically significant at n=6, consistent with the default-
OFF posture."
    ]
},
{
    "component": "Multi-signal fusion (shuttle + person + audio) cue ? feature producers, the
fusion_hybrid segmenter, and the offline experiment-harness scoring path",
    "present": true,
    "default_off": true,
    "no_special_casing": true,
    "retained": true,
    "evidence": [
        "PRESENT ? producers exist: backend/pipeline/detectors/fusion_features.py:194-215 defines
PERSON_FEATURE_NAMES (5 cols) + compute_person_features() returning exactly those keys;
backend/pipeline/detectors/audio_features.py:15 AUDIO_FEATURE_NAMES (3 cols) +
compute_audio_features(); distill_local.py:40 SHUTTLE_FEATURE_NAMES (5 cols).",
        "PRESENT ? served segmenter exists: backend/pipeline/segmenters/fusion_hybrid.py:39 class
FusionSegmenter(TrajectoryHybridSegmenter), registered at :154
segmenter_registry.register(FusionSegmenter). Producers wired in: _enrich_candidate_stats
persists net_crossings always (:85) and person features only when cue_flags.person (:88-89); GPU
person path is lazy (:110-111 'if self._person is None').",
        "PRESENT ? offline measurement path: backend/eval/fusion_golden.py (GPU person extract ->
*_golden_features.json), fusion_compare.py (P3 shuttle vs shuttle+person LOVO), fusion_audio.py
(P4 incremental audio LOVO). All defer to backend/eval/experiment.py::compare (no new scoring
math).",
        "DEFAULT_OFF ? config schema: backend/config/models.py:96 'cue_flags: dict[str, bool] =
Field(default_factory=dict)' (empty => person OFF); :99 min_crossings default 0; :102

```


min_players default 0; IndexingConfig.default_segmenter default 'yolo_hybrid' (:112) ? NOT 'fusion'.",

"DEFAULT_OFF ? on-disk config: config.json has default_segmenter='gemini' and NO 'fusion' block at all (verified via json load: \"'fusion' in indexing: False\"), so every fusion knob takes its default-OFF value. fusion_hybrid is opt-in only ? fusion_hybrid.py:22-26 docstring 'Registered as fusion_hybrid, NOT the default segmenter ... Promotion to default happens only on measured LOVO lift ... never silently.'",

"DEFAULT_OFF ? even when fusion_hybrid IS selected, default thresholds are no-ops: _select_for_handoff (fusion_hybrid.py:133-151) only drops windows when min_crossings>0 (:146) or person_on AND min_players>0 (:148); persisted candidates remain the full superset regardless (:138 comment).",

"NO_SPECIAL_CASING ? signals enter as feature COLUMNS the learned LogisticRegression weights: experiment.py:227-238 predict() builds _feature_matrix(vc, variant.feature_names) and filters by model.predict_proba >= threshold. The cue is added purely by extending Variant.feature_names: fusion_compare.py:59-61 fusion variant = SHUTTLE + PERSON names; fusion_audio.py:82-85 = SHUTTLE + PERSON + AUDIO names. No if/else per-cue branch in the decision.",

"NO_SPECIAL_CASING (one caveat) ? the ONLY hard if/else gates are net_crossings (Gate A) and players_in_court (Gate B) in experiment.py:_gate_mask (:200-210) and fusion_hybrid.py:_select_for_handoff (:146-148). These are explicit, opt-in, default-0 (OFF) decision GATES (rally_gate's net-crossing heuristic), distinct from the learned fusion columns; the person/audio FUSION cue itself is never a hard branch.",

"RETAINED ? measured n=6 ?F1 was non-significant: eval_baselines/experiment_leaderboard.json shows person ?F1=-0.021 CI[-0.165,+0.162] ci_excludes_zero=false (sign 2W/4L p=0.688); audio incr ?F1=+0.079 CI[-0.054,+0.203] ci_excludes_zero=false. docs/QUALITY_ITERATIONS.md:54 'neither cue clears the promotion bar ... at n=6 -> both stay default-OFF'; :59-60 'Negative/uncertain results are kept on purpose ? this is the honest ablation backbone for the paper aim.'",

"RETAINED ? nothing deleted: producer + feature math still present (segmenter_registry.register(FusionSegmenter) at fusion_hybrid.py:154 still there). The only fusion-code change after the n=6 measurement commit (#127) was experiment.py +122/-8 (#130, the additive C2 threshold-tuning/calibration fix) ? git diff --stat d7b1f51..HEAD shows no fusion_features.py/fusion_hybrid.py/fusion_compare.py/fusion_audio.py deletions. Re-measurable by re-running fusion_compare/fusion_audio on more golden videos.",

"FORWARD-LOOKING SEAM CONFIRMED (do-not-build) ? adding a 'shuttle-on-floor for a contiguous frame sequence' signal needs exactly the audio-cue pattern, zero special-casing: (1) NEW backend/pipeline/detectors/shuttle_floor_features.py with SHUTTLE_FLOOR_FEATURE_NAMES + compute_shuttle_floor_features(points/per_frame, ...) -> dict of those keys (mirrors fusion_features.py:194/200 and audio_features.py:15/18, both pure/no-IO/[0,1] floats). (2) In fusion_golden.extract_video (fusion_golden.py:108-115) add a 'shuttle_floor' sub-dict to each candidate exactly like the 'person' dict; OR follow fusion_audio.augment (fusion_audio.py:43-44) to add it post-hoc with no GPU. (3) A new Variant whose feature_names = SHUTTLE + PERSON + AUDIO + SHUTTLE_FLOOR_FEATURE_NAMES (extend the list at fusion_compare.py:60 / fusion_audio.py:84). The learned scorer weights it automatically via experiment.py:_feature_matrix (:195-197) ? and experiment.py:_feature_column (:180-192) already DEFAULTS a missing column to 0.0 with a warning, so old feature JSONs / DB rows that lack it still vectorise (graceful back-compat). (4) For SERVED use it would persist into candidate_segments.stats via a new branch in fusion_hybrid._enrich_candidate_stats gated by a new cue_flags entry (e.g. cue_flags.shuttle_floor), staying default-OFF like cue_flags.person at fusion_hybrid.py:88. No decision-path edit required to ADD the column."

],

"concerns": [

"The two served decision GATES (Gate A net_crossings, Gate B players_in_court) ARE hard-coded if/else thresholds in BOTH the offline path (experiment.py:_gate_mask :200-210) and production (fusion_hybrid.py:_select_for_handoff :146-148), not learned-scored. They are default-OFF (threshold 0) and explicitly the rally_gate heuristic, so they don't violate the no-special-casing claim for the FUSION cue ? but a future shuttle-floor signal could be mis-implemented as a third such hard gate (e.g. a min_floor_frames knob) instead of a weighted feature column. The clean architecture is the feature-column path; flag this so the future signal is added to feature_names, not as another _gate_mask branch.",

"Gate A (net_crossings) was actually PROMOTED to a recommended config (min_crossings=2) per QUALITY_ITERATIONS.md:62-67, but config.json still has NO fusion block, so it is not active in the shipped served path ? only documented as 'recommended'. Not a default-OFF violation, but the doc 'promoted' wording slightly overstates shipped state vs. config.json reality.",

"The person feature producer's REAL-VIDEO path (fusion_hybrid.py:_person_features and

PersonDetector.detections_over_windows) is only mock-tested (fusion_hybrid.py:108 '? real-video verification is owner-side; this wiring is mock-tested') and cannot run in this GPU-free container. present/no-special-casing are verified by code reading; the actual GPU feature values feeding the scorer are unverified here.",

"Audio (P4) ?F1 is measured INCREMENTALLY on top of person, never on shuttle-only directly (fusion_audio.py:79-86), so the audio leaderboard number carries the person 'handicap' (QUALITY_ITERATIONS.md:55-58 notes this and lists it as next work). Doesn't affect retention/default-OFF, but the +0.079 figure is not audio's standalone lift.",

"Minor: there is also an older 'audio onset detector' that was evaluated and NOT adopted for fusion (per memory note audio-e1-findings); the current fusion_audio.py uses backend/pipeline/audio/hit_detector.detect_hits_in_video ? confirm these aren't conflated when raising N. Not a defect, just lineage to keep straight."

```
]
}
```

ADVERSARIAL:

```
{
  "refuted": false,
  "refutations": [
    "NO VIOLATION FOUND ? could not refute any of the four claims for any of the three weak signals (person, audio, court-bounds).",
    "Searched for a DELETED/absent signal: none. `git log --all --diff-filter=D` over person_detector.py, audio_features.py, fusion_features.py, hit_detector.py, fusion_audio.py, fusion_compare.py returns EMPTY. All 9 source files are git-tracked and on origin/master; every producer and feature-math function is intact.",
    "Searched for a cue ON by default: none. backend/config/models.py:96 cue_flags defaults `{}` (person OFF), :99/:102 min_crossings/min_players default 0 (gates OFF), :104 court_polygon default None, :112 default_segmenter='yolo_hybrid'. Shipped config.json loads with default_segmenter='gemini' and NO 'fusion' block at all (verified via json.load) ? so every fusion knob takes its default-OFF value. No fusion-family segmenter is a default.",
    "Searched for a hard if/else special-case in the LEARNED decision path: none. experiment.py:227-238 routes every cue through Variant.feature_names -> _feature_matrix -> model.predict_proba -> threshold. classifier.py:64 LogisticRegression learns one weight per column (self.w=np.zeros(d)) with zero per-cue branching. audio_features.py and fusion_features.py return pure float dicts, never a keep/drop. The ONLY hard if/else gates (_gate_mask experiment.py:200-210; _select_for_handoff fusion_hybrid.py:146-148) are the deliberate, separately-documented Gate A/B (net_crossings/players_in_court), both default-0 (OFF) and NOT the path by which a cue enters the learned scorer ? so the no-special-casing claim for the learned fusion holds.",
    "Searched for code that disables a cue BECAUSE its measured lift was weak: none. No disable/remove commit exists in history. docs/QUALITY_ITERATIONS.md:54-60 records person ?F1=-0.021 and audio ?F1=+0.079 (both CIs span 0, n=6) and explicitly states 'both stay default-OFF (promote-only-on-lift)' and 'Negative/uncertain results are kept on purpose ? this is the honest ablation backbone.' Code was retained, not deleted, precisely because the result was inconclusive.",
    "Searched for a place a weak signal is LOST as the corpus grows: the actual SCORER path is back-compatible ? experiment.py:185-188 _feature_column defaults a missing/absent column to 0.0 with a warning, so old feature JSONs that lack a column still vectorise. A new column is added by extending feature_names (fusion_compare.py:60, fusion_audio.py:84), not by editing the decision path. No signal is silently dropped from scoring."
  ],
  "confirmations": [
    "PERSON cue PRESENT + retained: fusion_features.py:194 PERSON_FEATURE_NAMES (5 cols) + :200 compute_person_features (pure, returns exactly those float keys); person_detector.py producer intact; both unchanged after the non-significant n=6 result.",
    "AUDIO cue PRESENT + retained: audio_features.py:15 AUDIO_FEATURE_NAMES (3 cols) + :18 compute_audio_features (pure numeric, 0.0 on empty window); hit_detector.py producer intact; fusion_audio.py eval driver intact.",
    "COURT-BOUNDS PRESENT + dormant-but-plumbed: fusion_features.py point_in_polygon/person_in_court/in_court_counts thread court_polygon; fusion_hybrid.py:126-131 wires fcfg['court_polygon'] into compute_person_features; FusionConfig.court_polygon default None (models.py:104) degrades to count-everyone. Catalyst fires when a polygon is supplied, no code change needed.",
    "DEFAULT-OFF verified at every layer: cue_flags={} , min_crossings=0, min_players=0,
```

```

court_polygon=None, default_segmenter='yolo_hybrid' (config.json ships 'gemini');
fusion_hybrid.py:88 person features only computed when cue_flags['person'] is True (lazy torch
import only then); fusion_hybrid.py:24-26 docstring 'NOT the default segmenter ? it is opt-in.
Promotion ... only on measured LOVO lift ... never silently.'",
    "NO-SPECIAL-CASING (weighted columns) verified: fusion_compare.py:57-61 and
fusion_audio.py:79-85 build variants differing ONLY by feature_names (SHUTTLE vs +PERSON vs
+PERSON+AUDIO); experiment.py:234 model.predict_proba(_feature_matrix(vc,
variant.feature_names)); shuttle's own net_crossings already rides as a learned column
(distill_local.py:40). LogisticRegression is fully cue-agnostic.",
    "RETAINED-not-deleted invariant test exists: tests/test_experiment.py:78-83
test_disabled_cues_byte_identical_to_control asserts a variant carrying only disabled cues
predicts identically to CONTROL ? the no-regression guarantee.",
    "PERSISTENCE invariant supports re-measurability: fusion_hybrid.py:19-20/:136-138 persisted
candidates stay the full superset regardless of served gating; fusion_golden writes replayable
*_golden_features.json so fusion_compare/fusion_audio can re-score any variant as N grows.",
    "ACKNOWLEDGED non-violating caveats (consistent with the audit): (a) the two served gates
min_crossings/min_players ARE hard if/else but are a deliberate default-OFF decision-gate (Gate
A/B), distinct from the learned-column path; (b) the real-video person/audio producer paths are
mock-tested only (GPU-free container) ? code/math verified by reading, GPU values owner-side;
(c) minor loader strictness inconsistency: fusion_compare.load_videos:47-48 and
fusion_audio.load_videos_with_audio:66-67 index c['person']/c['audio'] directly (would KeyError,
i.e. fail LOUDLY, on a future JSON missing the dict) whereas the scorer's _feature_column
defaults missing columns to 0.0 ? a fail-loud inconsistency, not a silently-lost-signal
violation; (d) the +0.079 audio number is an INCREMENTAL lift measured on top of the person cue,
not standalone-vs-shuttle (flagged as next work in QUALITY_ITERATIONS.md:58)."
```

Produce: (1) a one-line VERDICT on whether "the code is all there and weak signals are default-OFF + retained + no-special-casing" holds; (2) a markdown table: Component | Present | Default-OFF | No-special-casing | Retained | Key refs; (3) any GAPS that need a fix (or "none"); (4) a short "Shuttle-drop slot-in" paragraph stating exactly where a future shuttle-on-floor-contiguous feature plugs in. Be concrete with file refs. Keep it under ~250 words.

2. assistant (2026-06-14T06:37:42.915Z)

Fusion weak-signals audit ? owner report

****VERDICT:**** ? Holds ? all three weak signals (person, audio, court-bounds) are PRESENT, DEFAULT-OFF at every config layer, scored as learned LogisticRegression feature-columns (no special-casing in the learned path), and RETAINED unchanged after the non-significant n=6 ?F1; the only nit is one untracked baseline JSON.

Component	Present	Default-OFF	No-special-casing	Retained	Key refs
Person cue	?	?	?	?	`fusion_features.py:194,200`; `person_detector.py:50`; `models.py:96`; `fusion_hybrid.py:88`
Audio cue	?	?	(eval-only)	?	`audio_features.py:15,18`; `hit_detector.py:136`; `fusion_audio.py:84`
Court-bounds	?	(plumbed, dormant)	?	?	`fusion_features.py:30,67,200`; `fusion_hybrid.py:126-131`; `models.py:104`
FusionSegmenter + config	?	?	?	?	`fusion_hybrid.py:39,154`; `models.py:112`; `config.json:26`
Scoring seam (learned)	?	?	?	?	`experiment.py:227-238`; `classifier.py:40-79`; `distill_local.py:40`

****GAPS:****

- ****Untracked baseline**** ? `eval\_baselines/experiment\_leaderboard.json` exists on disk with both P3/P4 rows but is not git-tracked (sibling baselines are). Fix: `git add` + commit it.
- ****Branch divergence**** ? checkout is on `docs/i3-fusion-measured` (1 ahead/1 behind `origin/master`), not fast-forwardable. Re-cut from `origin/master` before any PR.
- Non-blocking: served person/audio path is mock-tested only (GPU-free container); the +0.079 audio lift is incremental-on-person, not standalone; court-polygon has no `[0,1]` validation. All by-design / owner-side.

****Shuttle-drop slot-in:**** A future "shuttle-on-floor-contiguous" cue plugs in with zero decision-path edits. (1) New `backend/pipeline/detectors/shuttle\_floor\_features.py` with `SHUTTLE\_FLOOR\_FEATURE\_NAMES` + a pure `compute\_shuttle\_floor\_features()` returning float keys (mirror `fusion\_features.py:194/200`, `audio\_features.py:15/18`). (2) Add a `shuttle\_floor` sub-dict per candidate in `fusion\_golden.py:108-115`, or post-hoc à la `fusion\_audio.augment` (`fusion\_audio.py:43-44`). (3) Extend a Variant's `feature\_names` (`fusion\_compare.py:60` / `fusion\_audio.py:84`) ? `experiment.py:\_feature\_matrix:195-197` weights it automatically, and `\_feature\_column:180-192` defaults missing columns to 0.0 (back-compat). (4) For served use, persist behind a new `cue\_flags['shuttle\_floor']` branch in `fusion\_hybrid.\_enrich\_candidate\_stats` (default-OFF like `:88`). Add it as a weighted column ? ****not**** a third `\_gate\_mask` hard threshold.